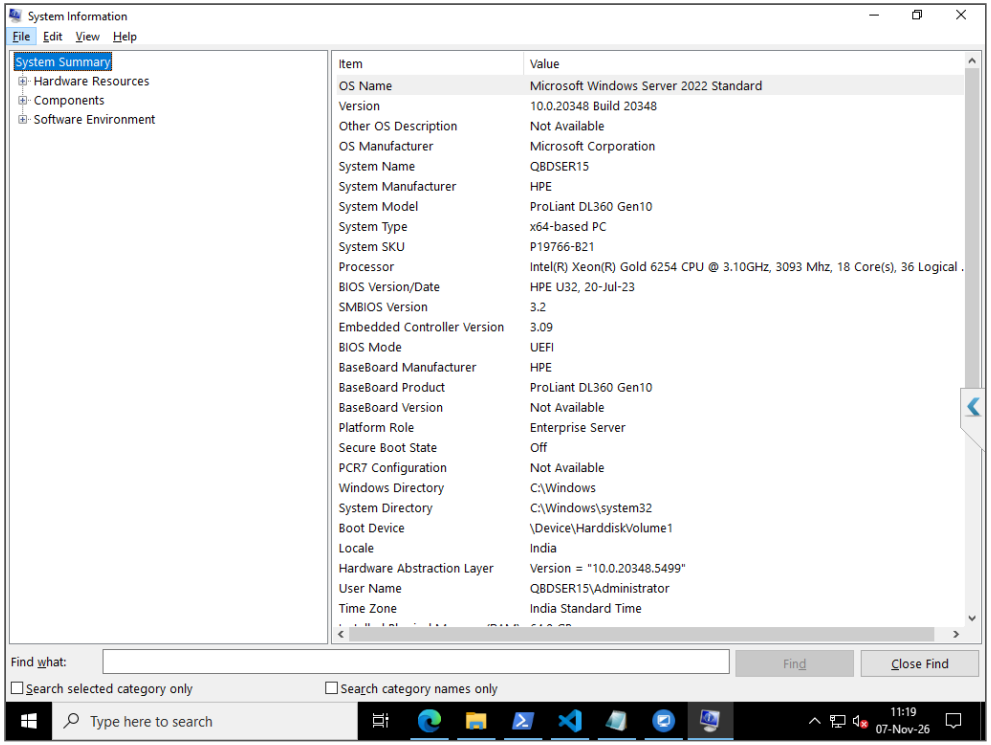
TCMS software Installation Qualification – Executed

INSTALLATION QUALIFICATION SCREENSHOTS FOR TCMS APPLICATION VER. 1.0

DOCUMENT NO.: LGC/QBD/0001	TITLE : COMPUTER SYSTEM VALIDATION OF TCMS

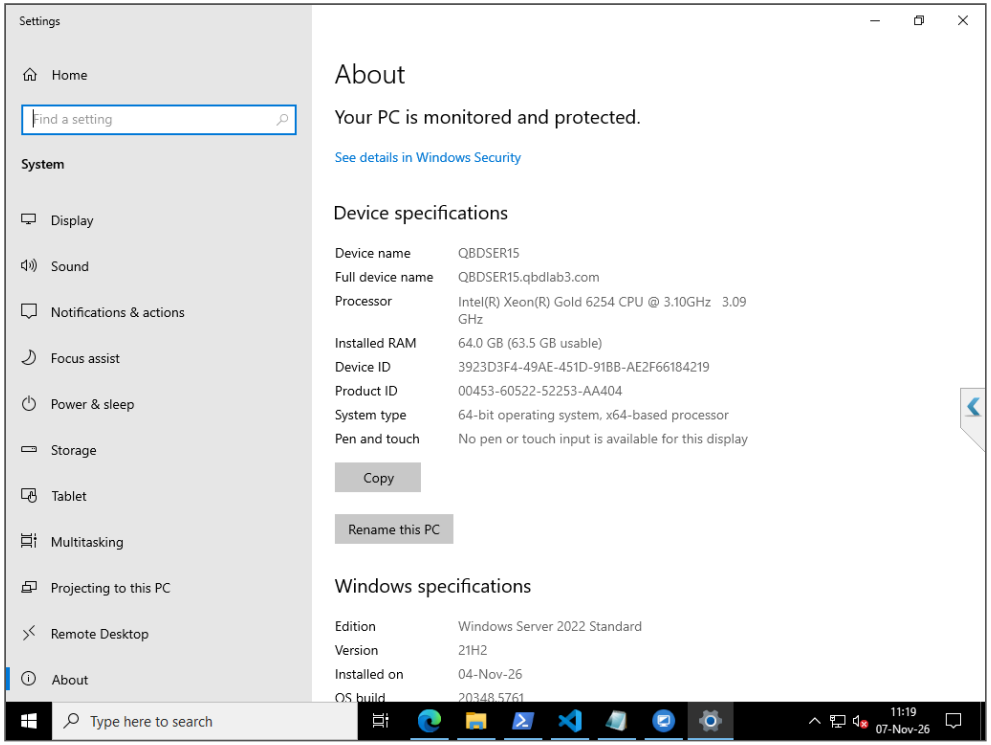
Test Objective :	To Verify Server Identification Details.			
Test Procedure :			**Expected Result :**	
Login to the operating system with admin credentials. Press Win + R, type msinfo32 and press Enter. In System Summary, verify System Name, System Manufacturer & System Model.			**1> System Name = QBDSER15 2> System Manufacturer = HPE 3> System Model = ProLiant DL360 Gen10**	
Actual Result :	1> System Name = QBDSER15 2> System Manufacturer = HPE 3> System Model = ProLiant DL360 Gen10			
Test ID : 8.1.1	**Step No : 1.1**	**Attachment No. : 1**	**Result : Pass**	**Executed By : _____ Date & Time : 07-Nov-26 11:19**



INSTALLATION QUALIFICATION SCREENSHOTS FOR TCMS APPLICATION VER. 1.0

DOCUMENT NO.: LGC/QBD/0001	TITLE : COMPUTER SYSTEM VALIDATION OF TCMS

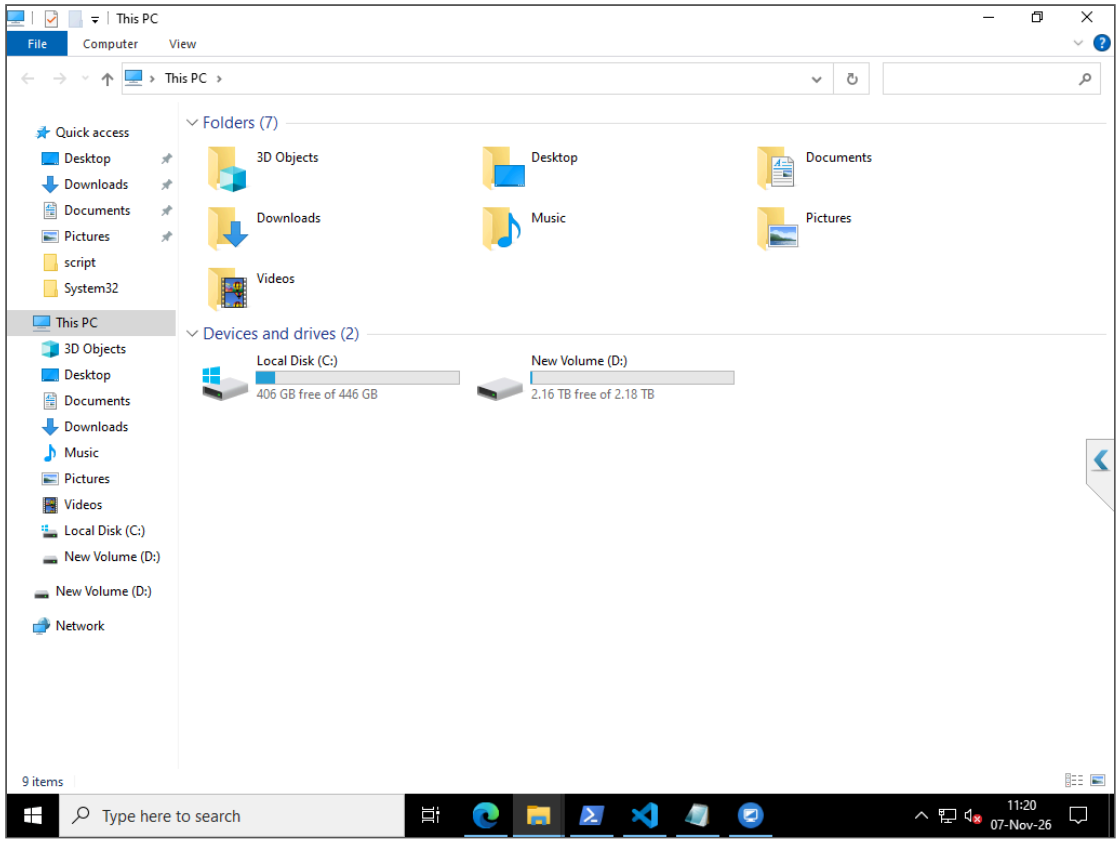
Test Objective :	To Verify Specification of Server Components and Software Components.			
Test Procedure :			**Expected Result :**	
Right Click on This PC from Desktop. Select Properties (Settings > System > About) and record the Processor, Installed Memory (RAM) & Operating System edition.			**1> Processor = Intel(R) Xeon(R) Gold 6254 CPU @ 3.10GHz 3.09 GHz 2> Installed Memory (RAM) = 64 GB and above 3> Operating System = Windows Server 2022 Standard**	
Actual Result :	1> Processor = Intel(R) Xeon(R) Gold 6254 CPU @ 3.10GHz 3.09 GHz 2> Installed RAM = 64.0 GB (63.5 GB usable) 3> Edition = Windows Server 2022 Standard (21H2)			
Test ID : 8.1.2	**Step No : 1.1**	**Attachment No. : 2**	**Result : Pass**	**Executed By : _____ Date & Time : 07-Nov-26 11:19**



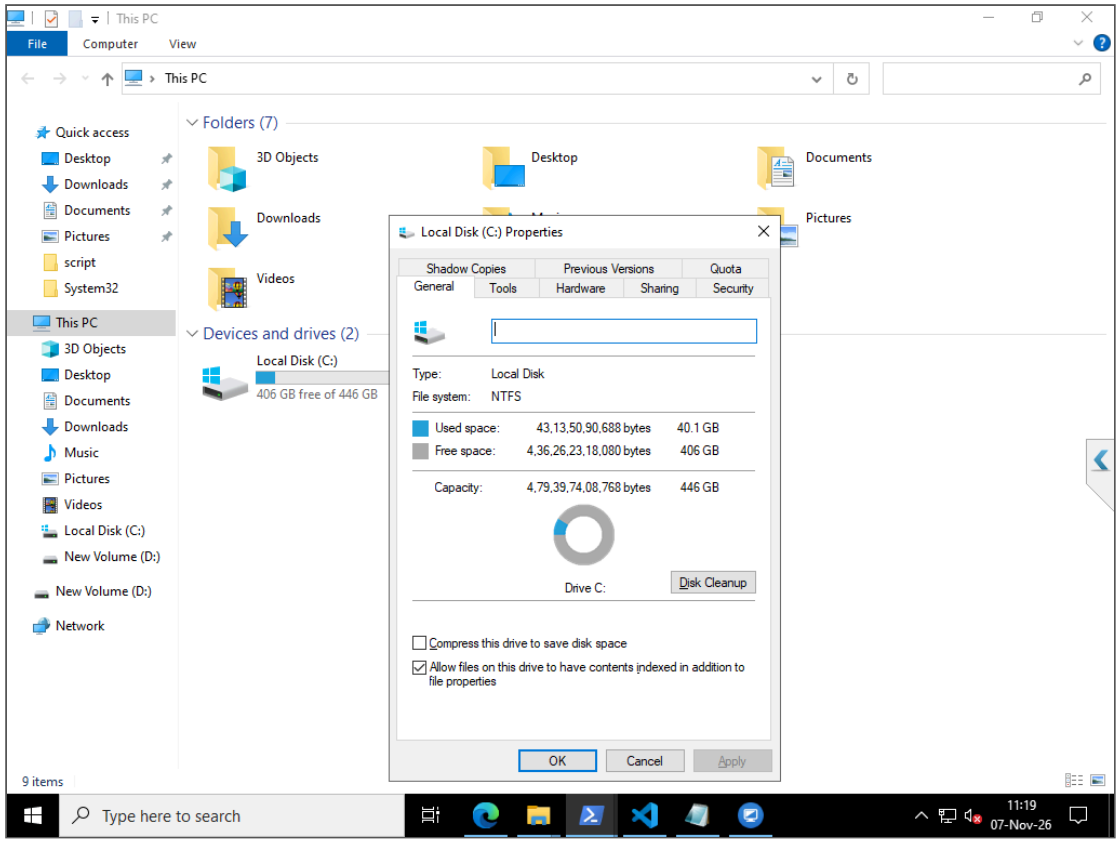
INSTALLATION QUALIFICATION SCREENSHOTS FOR TCMS APPLICATION VER. 1.0

DOCUMENT NO.: LGC/QBD/0001	TITLE : COMPUTER SYSTEM VALIDATION OF TCMS

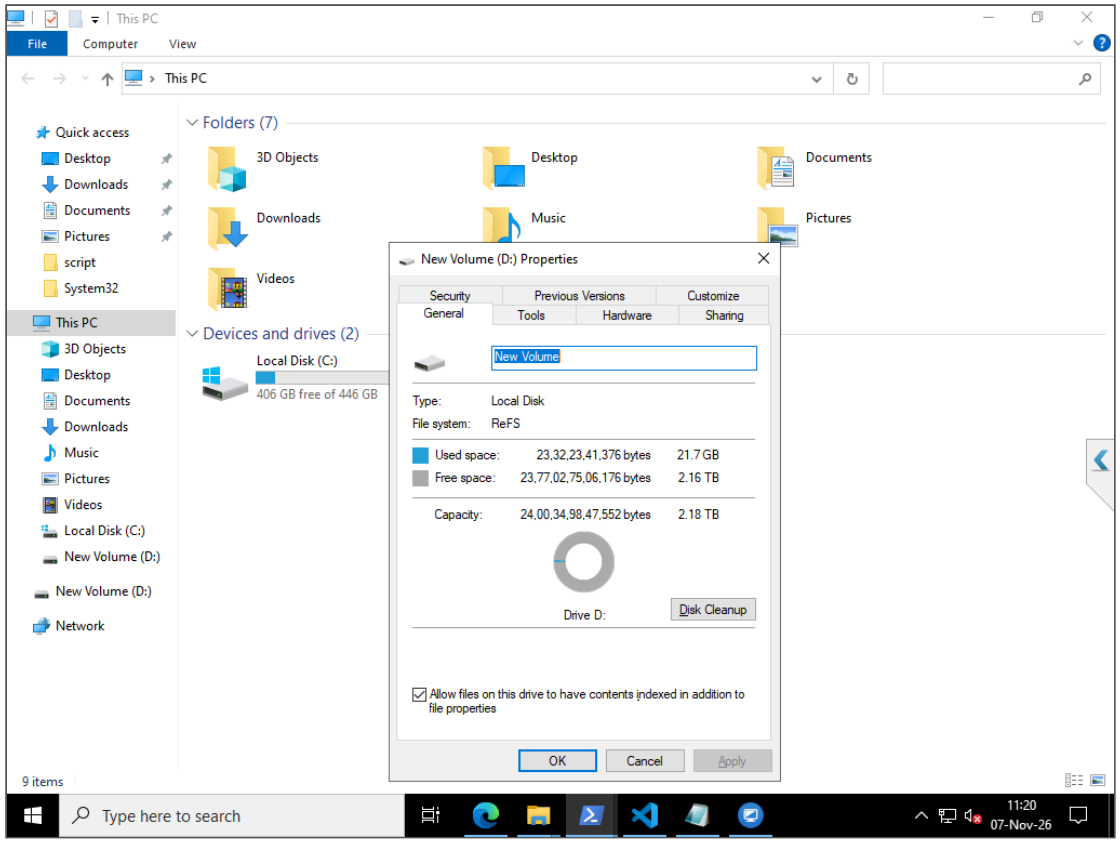
Test Objective :	To Verify Specification of Server Components and Software Components.			
Test Procedure :			**Expected Result :**	
Double click on This PC & record the Disk partitions. Right click on each drive, select Properties & record the Capacity.			**1> C:\ Capacity = 400 GB and Above 2> D:\ Capacity = 2 TB and Above**	
Actual Result :	Two drives listed: Local Disk (C:) 446 GB, New Volume (D:) 2.18 TB			
Test ID : 8.1.2	**Step No : 2.1**	**Attachment No. : 3**	**Result : Pass**	**Executed By : _____ Date & Time : 07-Nov-26 11:20**



Test Objective :	To Verify Specification of Server Components and Software Components.			
Test Procedure :			Expected Result :	
Double click on This PC & record the Disk partitions. Right click on each drive, select Properties & record the Capacity.			1> C:\ Capacity = 400 GB and Above 2> D:\ Capacity = 2 TB and Above	
Actual Result :	C:\ Capacity = 446 GB (NTFS)			
Test ID : 8.1.2	Step No : 2.2	Attachment No. : 4	Result : Pass	Executed By : _____ Date & Time : 07-Nov-26 11:19



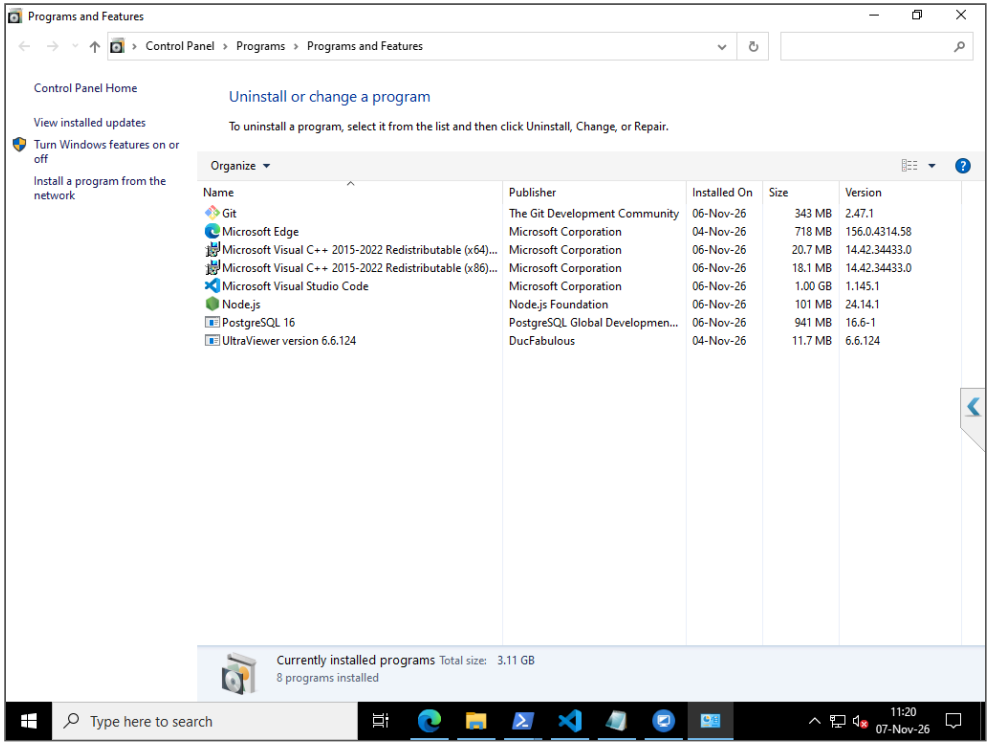
Test Objective :	To Verify Specification of Server Components and Software Components.			
Test Procedure :			Expected Result :	
Double click on This PC & record the Disk partitions. Right click on each drive, select Properties & record the Capacity.			1> C:\ Capacity = 400 GB and Above 2> D:\ Capacity = 2 TB and Above	
Actual Result :	D:\ Capacity = 2.18 TB (ReFS)			
Test ID : 8.1.2	Step No : 2.3	Attachment No. : 5	Result : Pass	Executed By : _____ Date & Time : 07-Nov-26 11:20



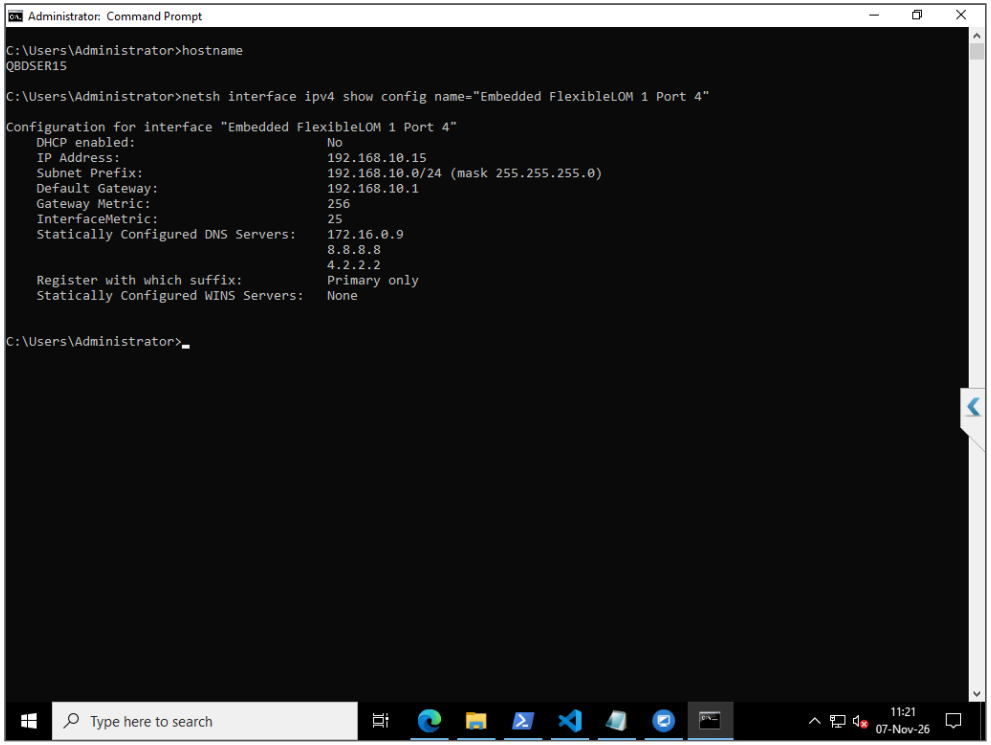
INSTALLATION QUALIFICATION SCREENSHOTS FOR TCMS APPLICATION VER. 1.0

DOCUMENT NO.: LGC/QBD/0001	TITLE : COMPUTER SYSTEM VALIDATION OF TCMS

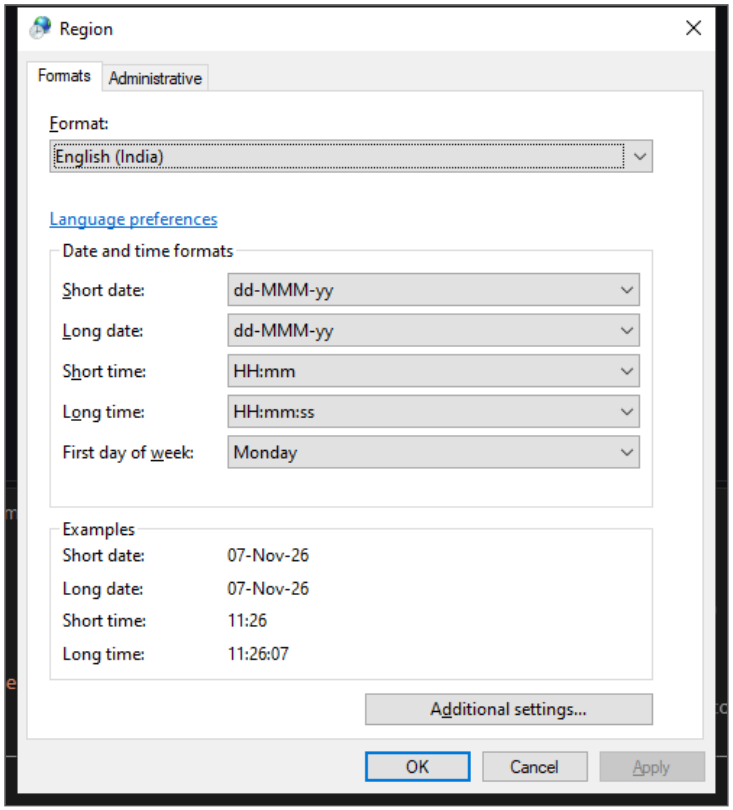
Test Objective :	To Verify Specification of Server Components and Software Components.			
Test Procedure :			**Expected Result :**	
Open Control Panel > Programs and Features & record the Installed Applications.			**1> Node.js 24.14.1 2> PostgreSQL 16 (16.6-1) 3> Microsoft Edge**	
Actual Result :	1> Node.js 24.14.1 2> PostgreSQL 16 - 16.6-1 3> Microsoft Edge 156.0.4314.58			
Test ID : 8.1.2	**Step No : 3.1**	**Attachment No. : 6**	**Result : Pass**	**Executed By : _____ Date & Time : 07-Nov-26 11:20**



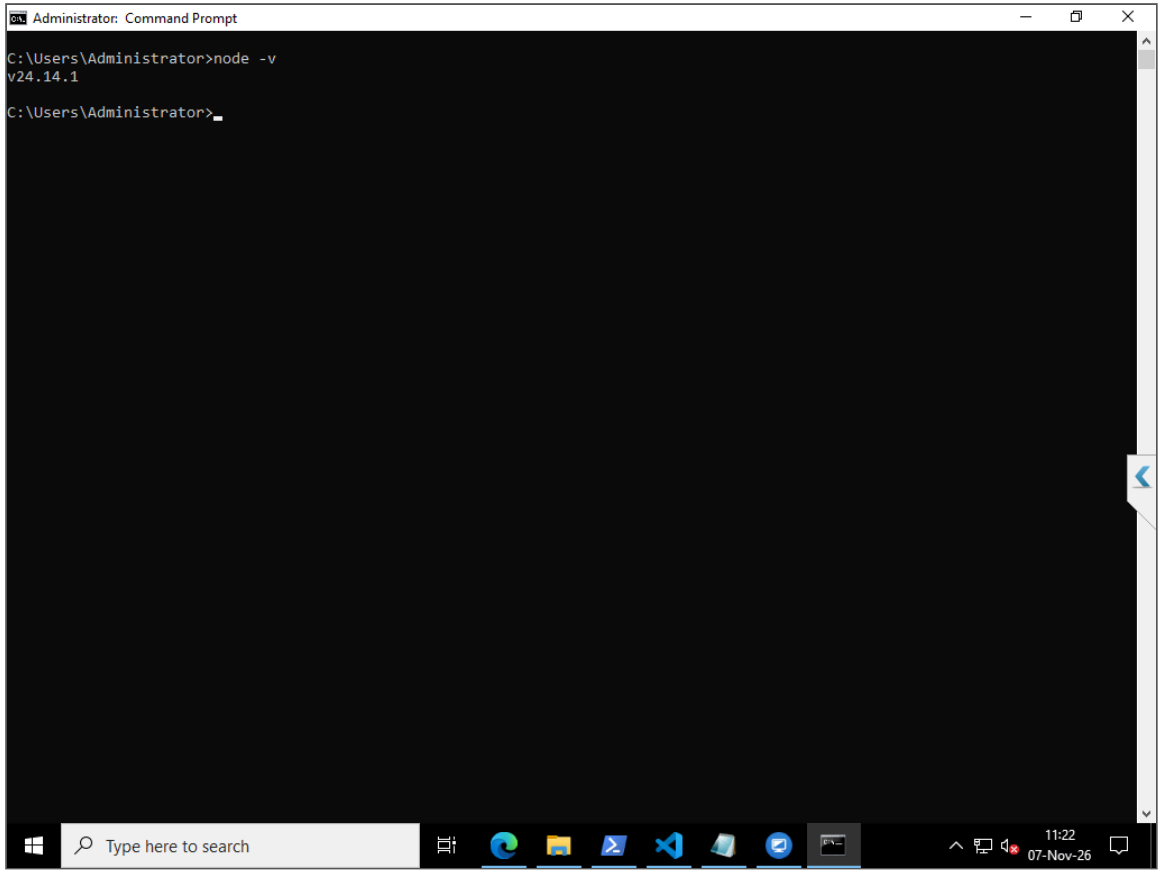
Test Objective :	**To verify the system network connectivity, System Date and time format.**			
Test Procedure :			**Expected Result :**	
Open Command Prompt as Administrator. Execute: hostname Execute: netsh interface ipv4 show config name="Embedded FlexibleLOM 1 Port 4"			**1> IP Address = 192.168.10.15 (DHCP enabled = No, i.e. Static) 2> Subnet mask = 255.255.255.0 3> Default Gateway = 192.168.10.1**	
Actual Result :	1> Host name = QBDSER15 2> IP Address = 192.168.10.15, DHCP enabled = No 3> Mask 255.255.255.0, Gateway 192.168.10.1			
Test ID : 8.1.3	**Step No : 1.1**	**Attachment No. : 7**	**Result : Pass**	**Executed By : _____ Date & Time : 07-Nov-26 11:21**



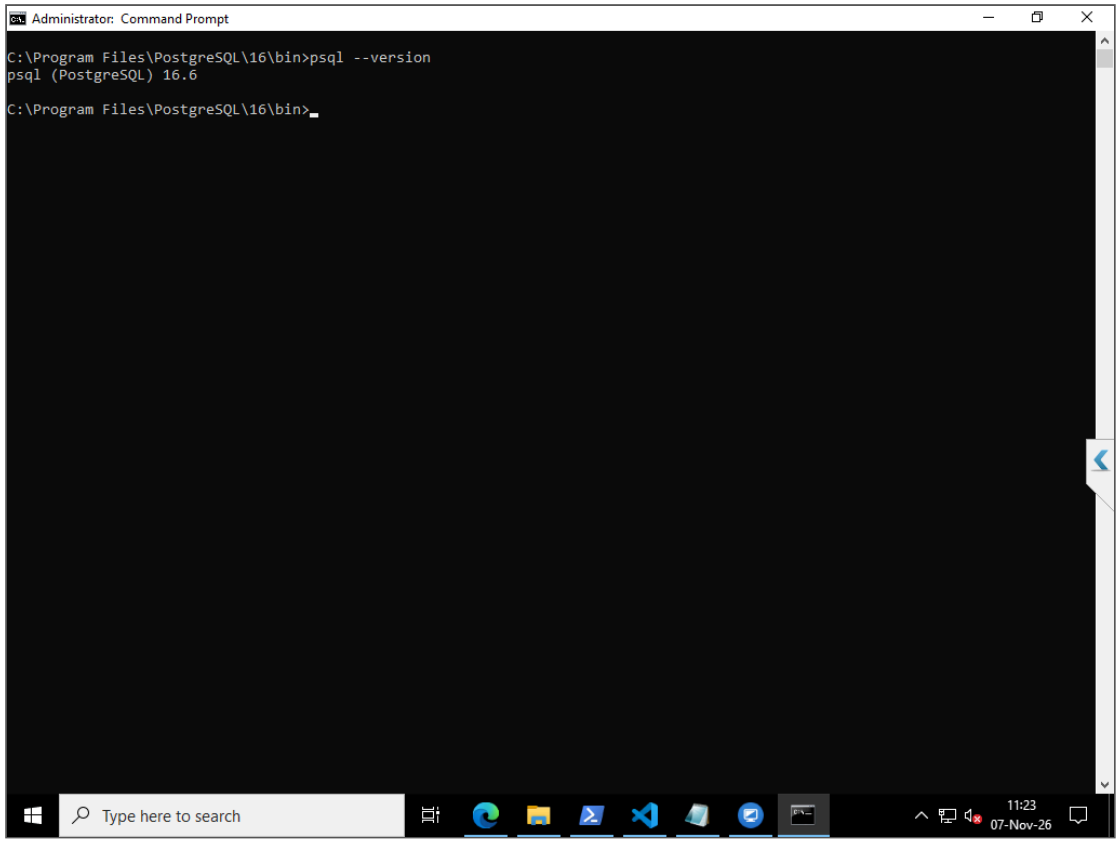
Test Objective :	**To verify the system network connectivity, System Date and time format.**			
Test Procedure :			**Expected Result :**	
Open Control Panel > Region (intl.cpl) and verify the date and time format.			**1> Short date format = dd-MMM-yy (e.g. 07-Nov-26) 2> Time format = HH:mm (24 Hrs.)**	
Actual Result :	1> Short date = dd-MMM-yy (07-Nov-26) 2> Short time = HH:mm, Long time = HH:mm:ss (24 Hrs.)			
Test ID : 8.1.3	**Step No : 2.1**	**Attachment No. : 8**	**Result : Pass**	**Executed By : _____ Date & Time : 07-Nov-26 11:26**



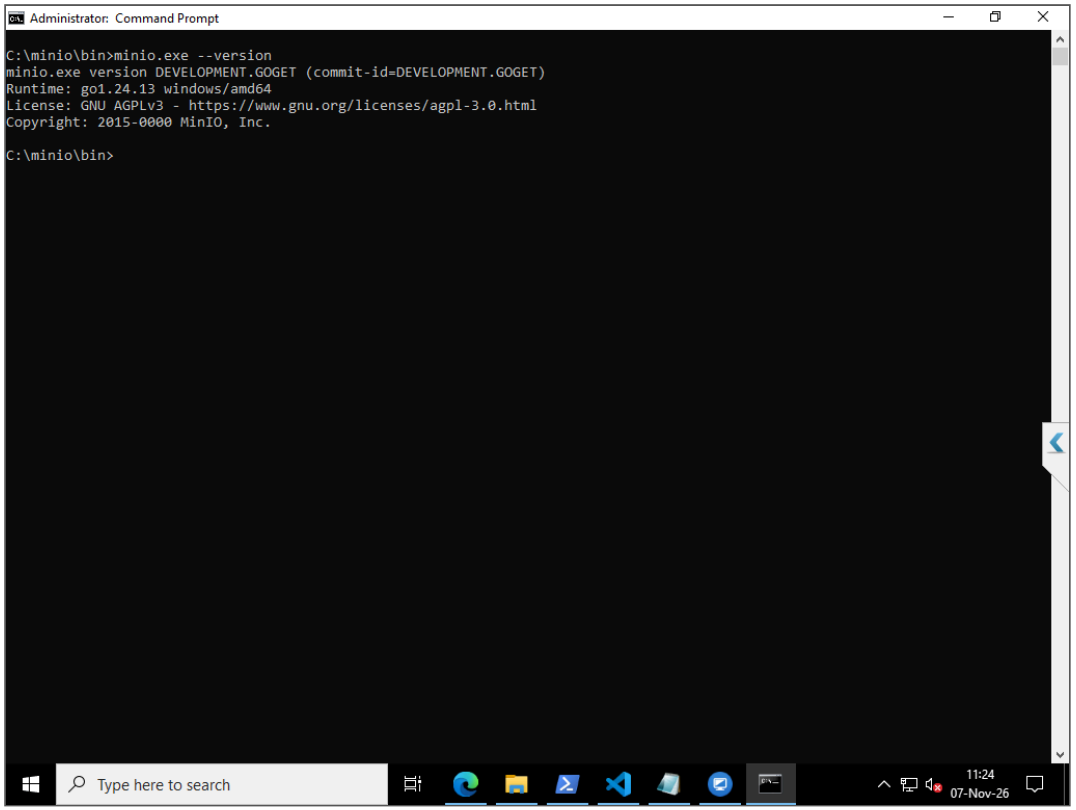
Test Objective :	To verify the versions of Node.js runtime, PostgreSQL, MinIO and NSSM installed for TCMS.			
Test Procedure :			Expected Result :	
Open Command Prompt as Administrator. Execute the command: node -v			Node.js version = v24.14.1	
Actual Result :	v24.14.1			
Test ID : 8.1.4	Step No : 1.1	Attachment No. : 9	Result : Pass	Executed By : _____ Date & Time : 07-Nov-26 11:22



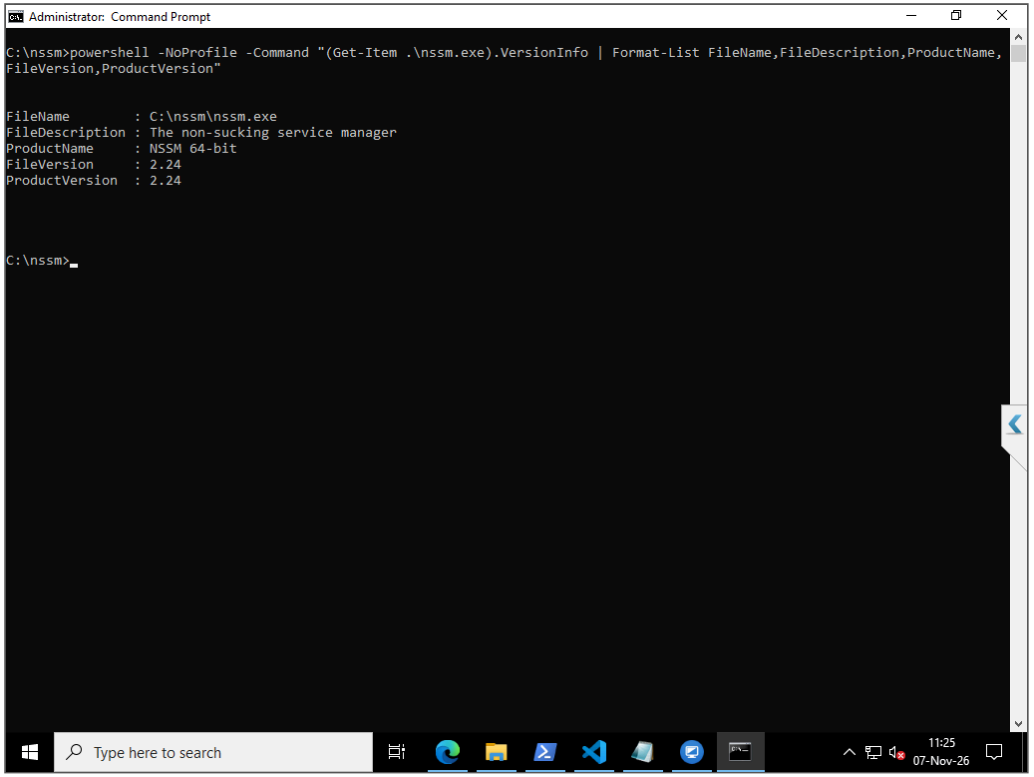
Test Objective :	To verify the versions of Node.js runtime, PostgreSQL, MinIO and NSSM installed for TCMS.			
Test Procedure :			Expected Result :	
In Command Prompt, navigate to C:\Program Files\PostgreSQL\16\bin. Execute the command: psql --version			psql (PostgreSQL) 16.6	
Actual Result :	psql (PostgreSQL) 16.6			
Test ID : 8.1.4	Step No : 2.1	Attachment No. : 10	Result : Pass	Executed By : _____ Date & Time : 07-Nov-26 11:23



Test Objective :	To verify the versions of Node.js runtime, PostgreSQL, MinIO and NSSM installed for TCMS.			
Test Procedure :			Expected Result :	
In Command Prompt, navigate to C:\minio\bin. Execute the command: minio.exe --version			MinIO server version is displayed. Record the version.	
Actual Result :	1> minio.exe version DEVELOPMENT.GOGET 2> Runtime: go1.24.13 windows/amd64			
Test ID : 8.1.4	Step No : 3.1	Attachment No. : 11	Result : Pass	Executed By : _____ Date & Time : 07-Nov-26 11:24



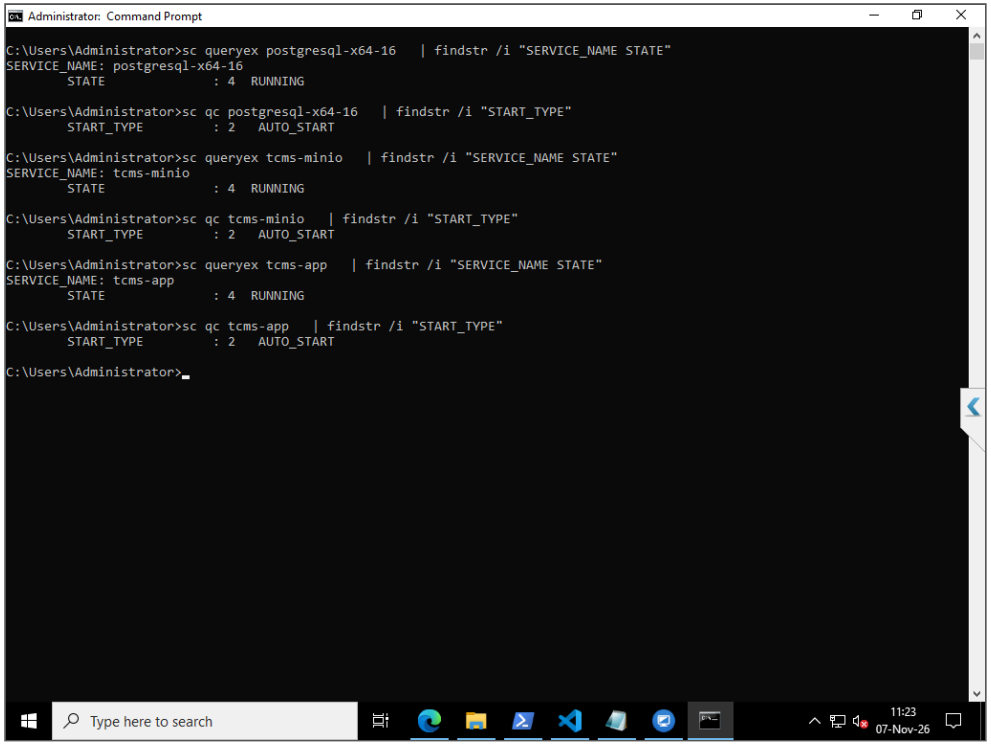
Test Objective :	To verify the versions of Node.js runtime, PostgreSQL, MinIO and NSSM installed for TCMS.			
Test Procedure :			**Expected Result :**	
In Command Prompt, navigate to C:\nssm. Execute: powershell -NoProfile -Command "(Get-Item .\nssm.exe).VersionInfo | Format-List FileName,FileDescription,ProductName,FileVersion,ProductVersion"			**1> ProductName = NSSM 64-bit 2> FileVersion = 2.24**	
Actual Result :	1> ProductName = NSSM 64-bit 2> FileVersion / ProductVersion = 2.24			
Test ID : 8.1.4	**Step No : 4.1**	**Attachment No. : 12**	**Result : Pass**	**Executed By : _____ Date & Time : 07-Nov-26 11:25**



INSTALLATION QUALIFICATION SCREENSHOTS FOR TCMS APPLICATION VER. 1.0

DOCUMENT NO.: LGC/QBD/0001	TITLE : COMPUTER SYSTEM VALIDATION OF TCMS

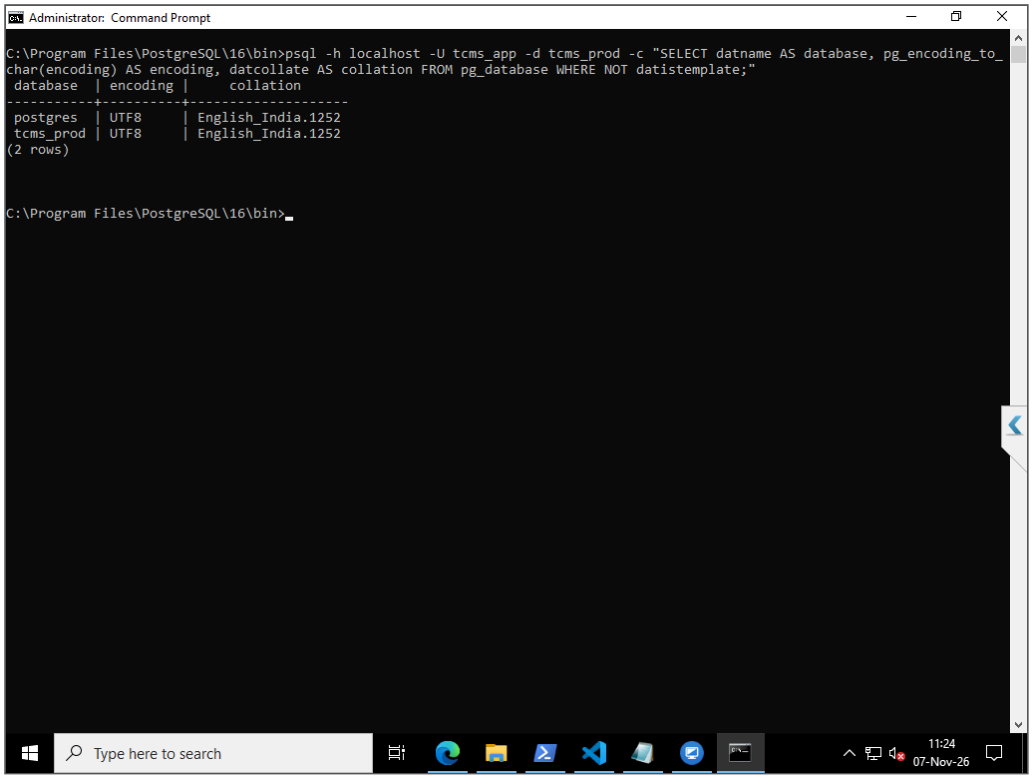
Test Objective :	**To verify TCMS components are installed as Windows services and running.**			
Test Procedure :			**Expected Result :**	
Open Command Prompt as Administrator. For each service (postgresql-x64-16, tcms-minio, tcms-app) execute:sc queryex <service> and sc qc <service>			**1> postgresql-x64-16: STATE = RUNNING, START_TYPE = AUTO_START 2> tcms-minio: STATE = RUNNING, START_TYPE = AUTO_START 3> tcms-app: STATE = RUNNING, START_TYPE = AUTO_START**	
Actual Result :	1> postgresql-x64-16: RUNNING, AUTO_START 2> tcms-minio: RUNNING, AUTO_START 3> tcms-app: RUNNING, AUTO_START			
Test ID : 8.1.5	**Step No : 1.1**	**Attachment No. : 13**	**Result : Pass**	**Executed By : _____ Date & Time : 07-Nov-26 11:23**



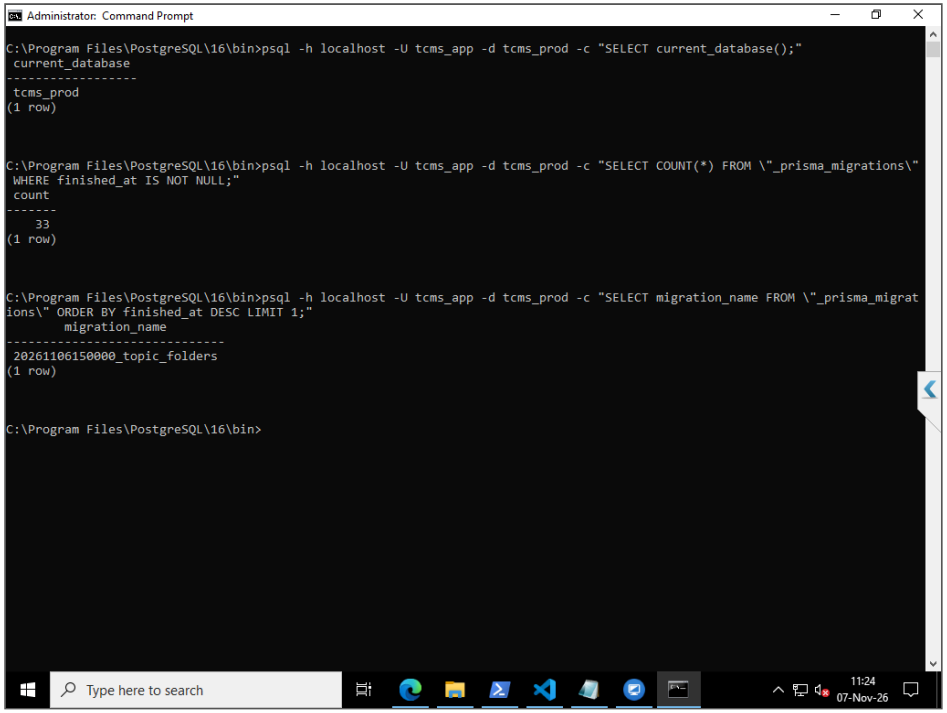
INSTALLATION QUALIFICATION SCREENSHOTS FOR TCMS APPLICATION VER. 1.0

DOCUMENT NO.: LGC/QBD/0001	TITLE : COMPUTER SYSTEM VALIDATION OF TCMS

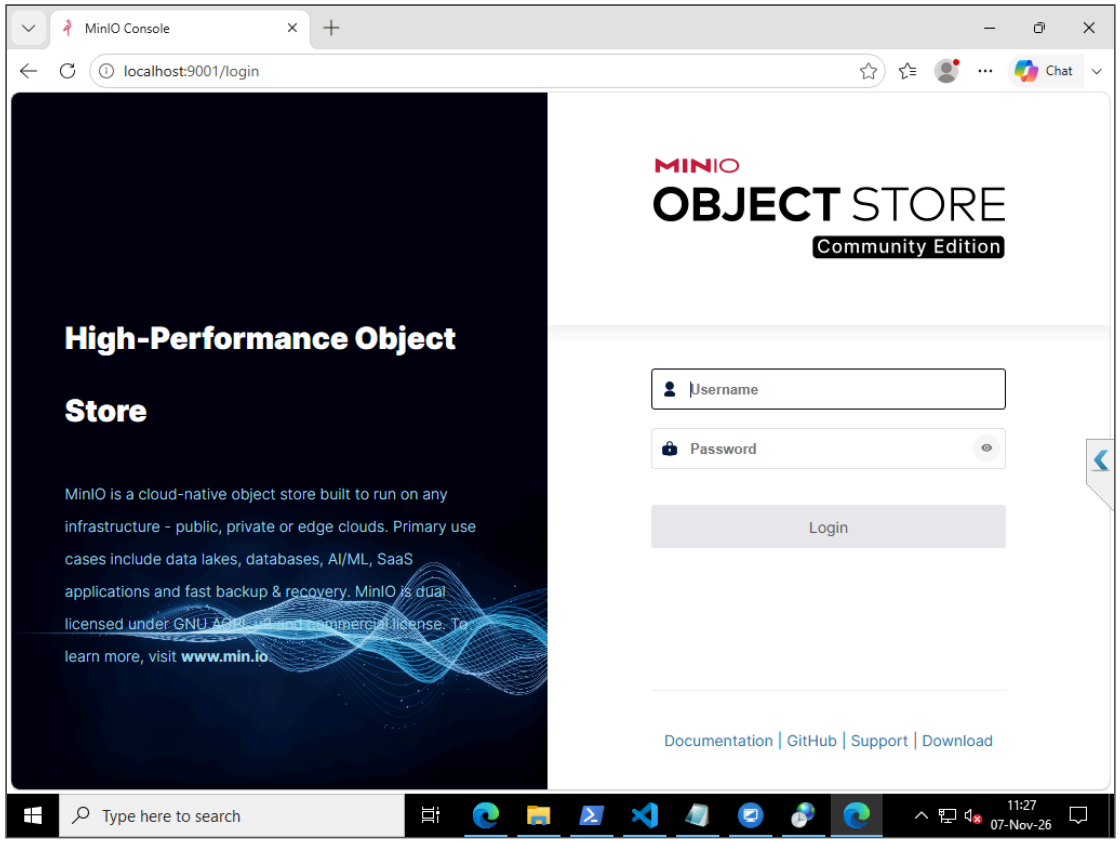
Test Objective :	To verify TCMS database creation and schema migrations.			
Test Procedure :			**Expected Result :**	
Open Command Prompt and navigate to C:\Program Files\PostgreSQL\16\bin. Execute: psql -h localhost -U tcms_app -d tcms_prod -c "SELECT datname AS database, pg_encoding_to_char(encoding) AS encoding, datcollate AS collation FROM pg_database WHERE NOT datistemplate;"			**Database tcms_prod is listed (UTF8).**	
Actual Result :	tcms_prod listed, encoding UTF8, collation English_India.1252			
Test ID : 8.1.6	**Step No : 1.1**	**Attachment No. : 14**	**Result : Pass**	**Executed By : _____** **Date & Time : 07-Nov-26 11:24**



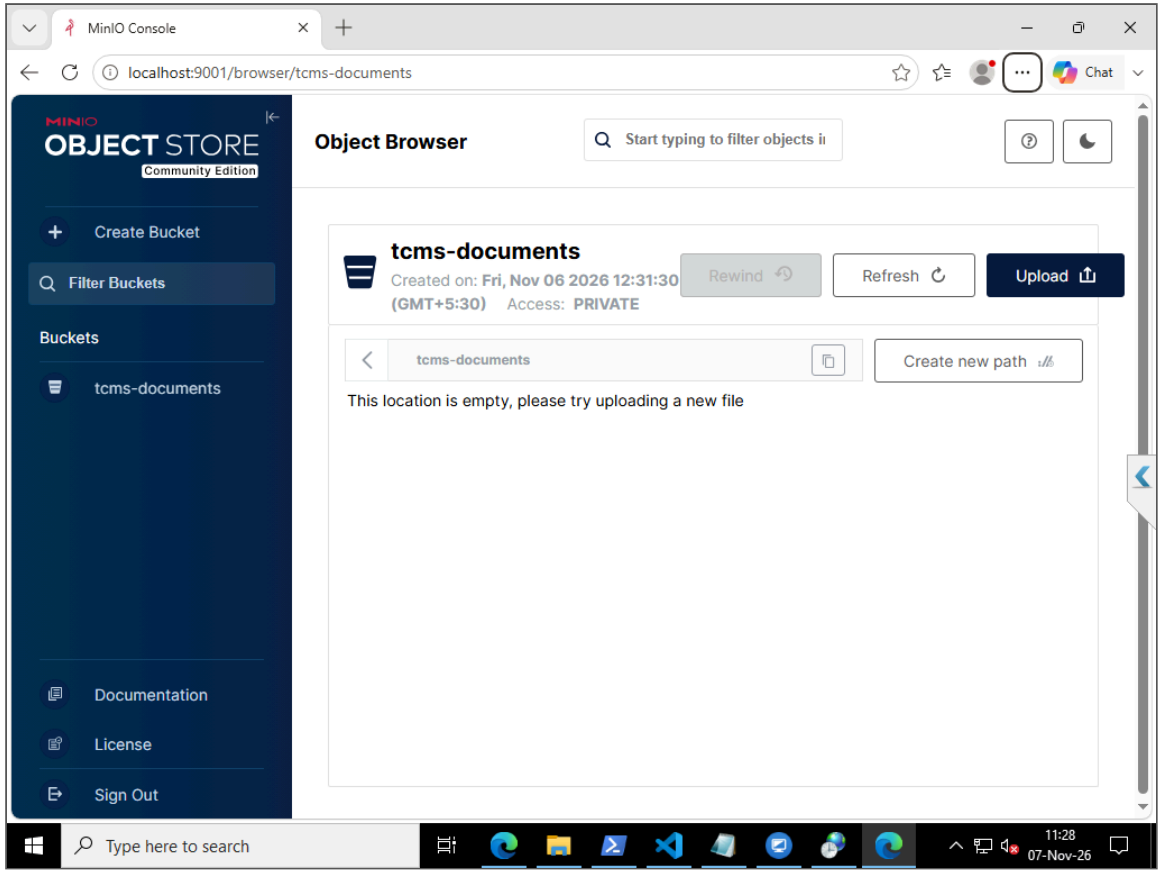
Test Objective :	To verify TCMS database creation and schema migrations.			
Test Procedure :			Expected Result :	
Execute: SELECT current_database(); Execute: SELECT COUNT(*) FROM "_prisma_migrations" WHERE finished_at IS NOT NULL; Execute: SELECT migration_name FROM "_prisma_migrations" ORDER BY finished_at DESC LIMIT 1;			1> Current database = tcms_prod 2> Count of applied migrations = 33 3> Latest migration = 20261106150000_topic_folders	
Actual Result :	1> current_database = tcms_prod 2> count = 33 3> migration_name = 20261106150000_topic_folders			
Test ID : 8.1.6	Step No : 2.1	Attachment No. : 15	Result : Pass	Executed By : _____ Date & Time : 07-Nov-26 11:24



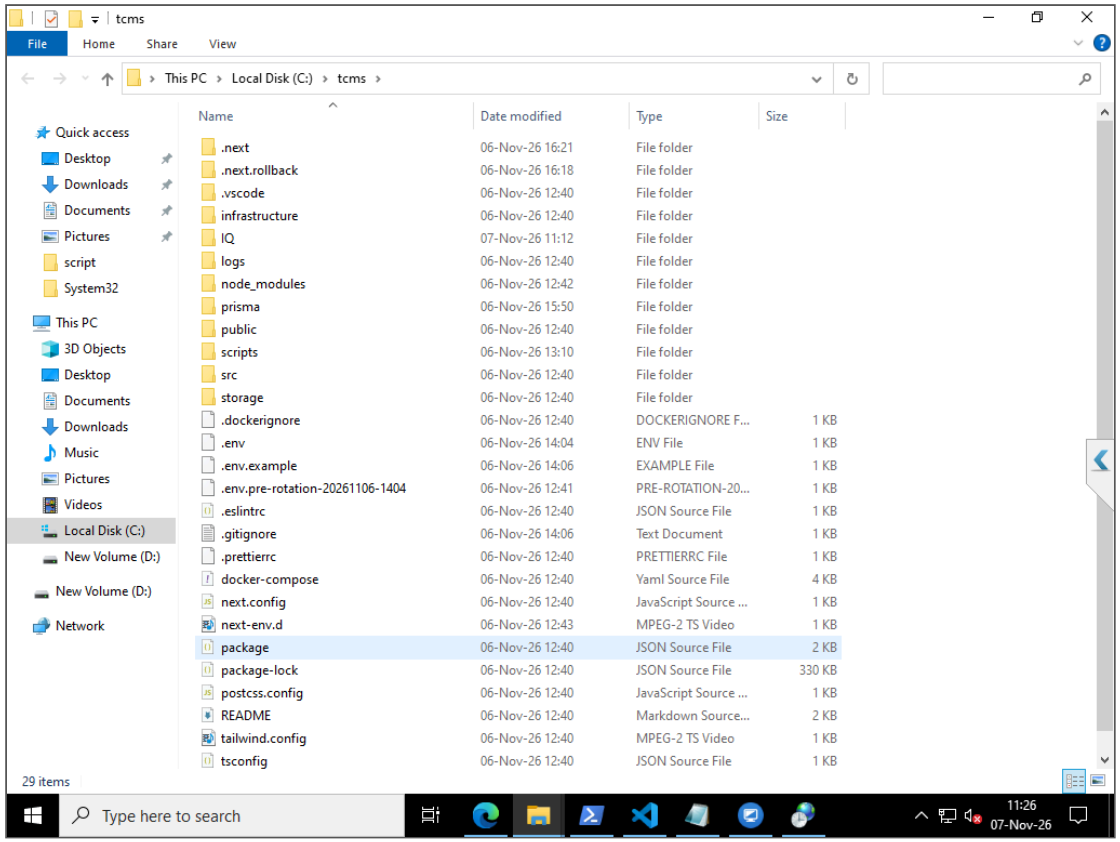
Test Objective :	To verify MinIO object storage and TCMS document bucket.			
Test Procedure :			Expected Result :	
On the server, open Microsoft Edge and navigate to http://localhost:9001. Login with MinIO administrator credentials.			MinIO Console login page is displayed and login is successful.	
Actual Result :	MinIO Console (Object Store - Community Edition) login page displayed; login successful (see Attachment 17).			
Test ID : 8.1.7	Step No : 1.1	Attachment No. : 16	Result : Pass	Executed By : _____ Date & Time : 07-Nov-26 11:27



Test Objective :	To verify MinIO object storage and TCMS document bucket.			
Test Procedure :			Expected Result :	
Open Object Browser / Buckets in the MinIO Console.			Bucket tcms-documents is listed.	
Actual Result :	Bucket tcms-documents listed (Access: PRIVATE, created 06-Nov-2026)			
Test ID : 8.1.7	Step No : 2.1	Attachment No. : 17	Result : Pass	Executed By : _____ Date & Time : 07-Nov-26 11:28



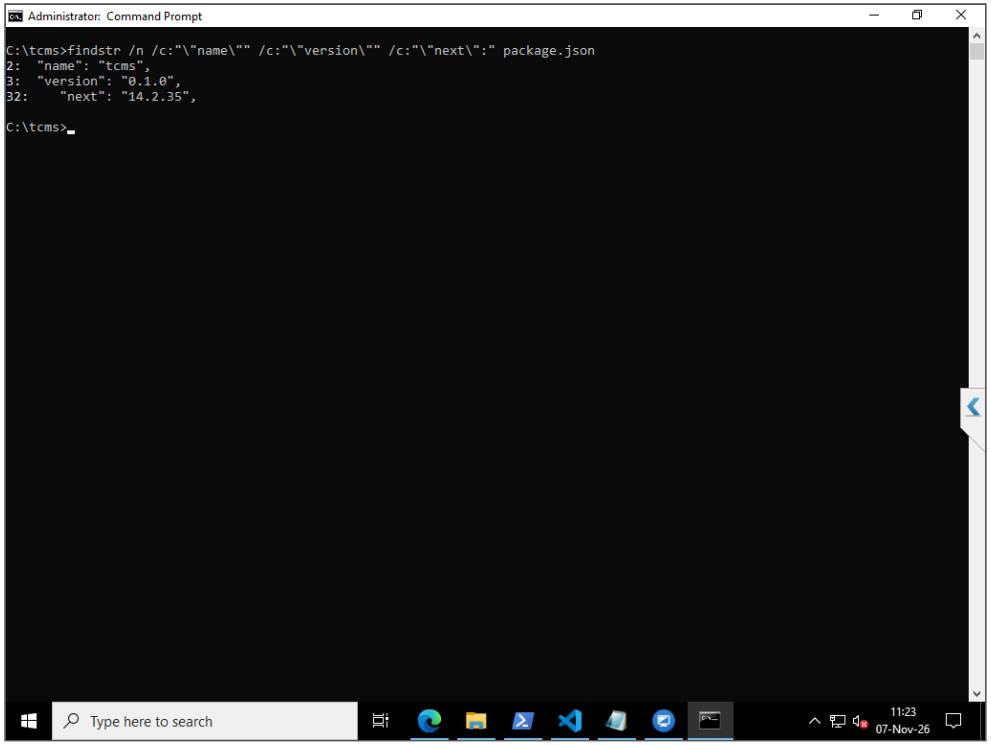
Test Objective :	To verify TCMS application installation files and configuration.			
Test Procedure :			Expected Result :	
Open File Explorer and navigate to the TCMS application installation directory C:\tcms. Record the installation path.			1> Installation path = C:\tcms 2> Directory contains: .next, node_modules, prisma, public, package.json and .env.	
Actual Result :	C:\tcms contains .next, node_modules, prisma, public, package.json, .env			
Test ID : 8.1.8	Step No : 1.1	Attachment No. : 18	Result : Pass	Executed By : _____ Date & Time : 07-Nov-26 11:26



INSTALLATION QUALIFICATION SCREENSHOTS FOR TCMS APPLICATION VER. 1.0

DOCUMENT NO.: LGC/QBD/0001	TITLE : COMPUTER SYSTEM VALIDATION OF TCMS

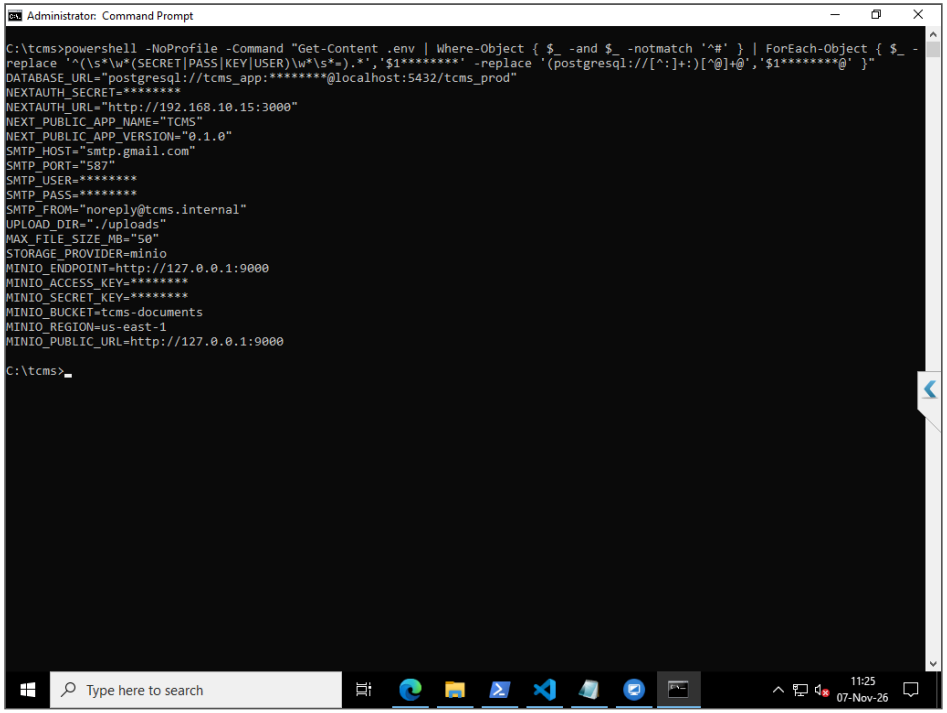
Test Objective :	To verify TCMS application installation files and configuration.			
Test Procedure :			**Expected Result :**	
In Command Prompt at C:\tcms, display the name, version and Next.js entries of package.json (read only).			**1> "name" = "tcms" 2> "version" = "0.1.0" 3> "next" = "14.2.35"**	
Actual Result :	1> "name": "tcms" 2> "version": "0.1.0" 3> "next": "14.2.35"			
Test ID : 8.1.8	**Step No : 2.1**	**Attachment No. : 19**	**Result : Pass**	**Executed By : _____ Date & Time : 07-Nov-26 11:23**



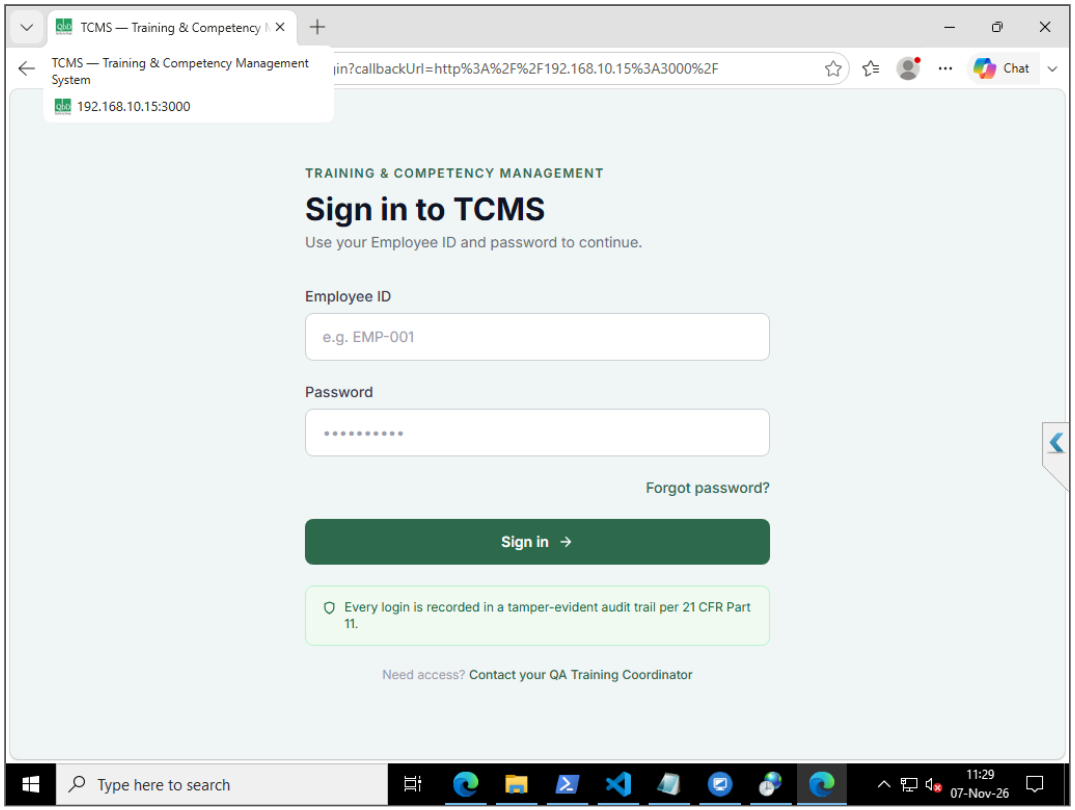
INSTALLATION QUALIFICATION SCREENSHOTS FOR TCMS APPLICATION VER. 1.0

DOCUMENT NO.: LGC/QBD/0001	TITLE : COMPUTER SYSTEM VALIDATION OF TCMS

Test Objective :	To verify TCMS application installation files and configuration.			
Test Procedure :			**Expected Result :**	
In Command Prompt at C:\tcms, display .env (read only) with all passwords / secret keys masked.			**1> NEXTAUTH_URL = http://192.168.10.15:3000 2> DATABASE_URL points to localhost:5432/tcms_prod 3> STORAGE_PROVIDER = minio 4> MINIO_BUCKET = tcms-documents**	
Actual Result :	1> NEXTAUTH_URL = "http://192.168.10.15:3000" 2> DATABASE_URL = postgresql://tcms_app:********@localhost:5432/tcms_prod 3> STORAGE_PROVIDER = minio, MINIO_BUCKET = tcms-documents			
Test ID : 8.1.8	**Step No : 3.1**	**Attachment No. : 20**	**Result : Pass**	**Executed By : _____ Date & Time : 07-Nov-26 11:25**



Test Objective :	**To verify TCMS application accessibility over the network.**			
Test Procedure :			**Expected Result :**	
Open Microsoft Edge. Navigate to http://192.168.10.15:3000 (server network address).			**1> TCMS Login page is displayed. 2> Browser tab title = "TCMS — Training & Competency Management System"**	
Actual Result :	1> "Sign in to TCMS" login page displayed 2> Tab title = "TCMS — Training & Competency Management System"			
Test ID : 8.1.9	**Step No : 1.1**	**Attachment No. : 21**	**Result : Pass**	**Executed By : _____ Date & Time : 07-Nov-26 11:29**



IQ Execution Summary

Sr. No.	Test ID	Test Title	Attachment No(s).	Result (Pass / Fail)	Discrepancy No. (if any)	Verified By (Sign & Date)
1.	8.1.1	Server Identification	1	**Pass**	NA	
2.	8.1.2	Server Components and Software Components.	2 – 6	**Pass**	NA	
3.	8.1.3	Server Network Connection & Date and Time Format	7 – 8	**Pass**	NA	
4.	8.1.4	Runtime and Supporting Software Version Verification	9 – 12	**Pass**	NA	
5.	8.1.5	Windows Services Verification	13	**Pass**	NA	
6.	8.1.6	Database Installation Verification	14 – 15	**Pass**	NA	
7.	8.1.7	Object Storage (MinIO) Verification	16 – 17	**Pass**	NA	
8.	8.1.8	Application Installation and Configuration Verification	18 – 20	**Pass**	NA	
9.	8.1.9	Application Accessibility Verification	21	**Pass**	NA	

Conclusion: Based on the above executed Installation Qualification scripts, the TCMS Application Ver. 1.0 has been installed on server QBDSER15 as per pre-approved specifications: Yes [✓] / No [ ]

Remarks:

1. All 21 screenshots were captured on server QBDSER15 on 07-Nov-26 (IST) as objective evidence. All passwords / secret keys were masked before capture.
2. Observation (8.1.4, Attachment 11): the installed MinIO binary reports version "DEVELOPMENT.GOGET" (Go runtime go1.24.13) instead of an official RELEASE tag. Expected result (version displayed and recorded) is met; replacement with an official MinIO release binary is recommended.
3. Observation (8.1.9): application accessibility was verified in Microsoft Edge using the server network address http://192.168.10.15:3000. Google Chrome is not installed on the server; Microsoft Edge is the qualified browser.

Approval

Prepared by:	Designation	Sign & Date
	(Logicology)	
Reviewed by:	**Designation**	**Sign & Date**
	(Logicology)	
	(QbD Research and Development Lab Pvt. Ltd.)	
Approved by:	**Designation**	**Sign & Date**
	(QbD Research and Development Lab Pvt. Ltd.)	
Signature indicates that the information above is accurate and the validation activities proposed are in compliance with regulatory and corporate requirements.		